

DeFi/Risk Analytics MVP

4-Week Detailed Implementation Plan

Executive Summary

Timeline:	4 weeks (January 30 - February 26, 2026)
Daily Commitment:	3 hours (excluding Sundays & Wednesdays)
Total Hours:	60 hours across 20 working days
Primary Focus:	DeFi/Risk Analytics Hybrid Discovery System
Key Deliverable:	Working demo + paper draft + 3 pilot customers

Strategic Context

Based on the Product Decision Analysis, we are starting with DeFi/Risk Analytics because:

- 280 DeFi queries and risk data already built (unfair advantage)
- Sharp target customer clarity: DeFi protocols need daily risk monitoring
- Higher novelty: AI-powered symbolic regression for DeFi formulas
- Faster time to revenue: 3-6 months vs 9-18 months for ANREPORT
- Growing market: DeFi job postings up 50%+ YoY
- Strong moat: Domain expertise + AI + symbolic reasoning

Work Schedule Overview

Working Days: Monday, Tuesday, Thursday, Friday, Saturday

Rest Days: Sunday, Wednesday

Daily Time: 3 hours per day

Week Total: 15 hours per week

Week 1: Market Validation & Core Architecture

Dates: January 30 - February 1

Working Days: Thursday, Friday, Saturday

Total Hours: 9 hours

Objective

Validate market need and establish technical foundation.

Day 1 - Thursday, January 30 (3 hours)

Focus: Market Research & Customer Discovery

Hour 1 (9:00-10:00): Research DeFi protocols needing risk analytics

- List top 20 DeFi protocols by TVL (Aave, Compound, Uniswap V4, Curve, etc.)
- Identify key decision-makers (Risk teams, Protocol economists)
- Research current risk management approaches (Gauntlet, Chaos Labs, in-house)

Hour 2 (10:00-11:00): Prepare outreach materials

- Draft cold email template highlighting 280 DeFi queries advantage
- Create 1-page problem statement: "Risk analytics takes weeks, we do it in hours"
- Prepare 3 questions for discovery calls

Hour 3 (11:00-12:00): Initial outreach

- Send 10 personalized emails to DeFi protocol risk teams
- Post in DeFi risk forums (Risk DAO, Gauntlet Discord)
- Schedule goal: 3-5 discovery calls by end of week

 **Deliverable:** 10 protocols contacted, outreach materials ready

Day 2 - Friday, January 31 (3 hours)

Focus: Technical Architecture Design

Hour 1 (9:00-10:00): System architecture planning

- Design 4-component architecture: Discovery → Validation → Interpretation → Output
- Define data flow between symbolic regression (PySR) and LLM (Claude)
- Identify external dependencies and API requirements

Hour 2 (10:00-11:00): Technology stack decisions

- Select: Python 3.11+, PySR for symbolic regression, Anthropic Claude API
- Choose validation frameworks: SymPy (symbolic), Pint (dimensional)
- Define data storage approach for 280+ formulas

Hour 3 (11:00-12:00): Create development roadmap

- Break MVP into 8 milestones across 4 weeks
- Identify critical path dependencies
- Set up project tracking (Notion/Linear/GitHub Projects)

✓ **Deliverable:** Architecture diagram, tech stack finalized, 4-week roadmap

Day 3 - Saturday, February 1 (3 hours)

Focus: Environment Setup & Data Preparation

Hour 1 (9:00-10:00): Development environment

- Set up Python virtual environment: `python -m venv venv`
- Install core packages: `pip install pysr anthropic sympy pint numpy pandas`
- Configure Anthropic API key and test connection

Hour 2 (10:00-11:00): Organize existing DeFi queries

- Review and categorize 280 DeFi queries by type (IL, AMM, liquidations, etc.)
- Identify top 30 priority formulas for MVP
- Create test data structure for validation

Hour 3 (11:00-12:00): Create project structure

- Initialize Git repository with proper `.gitignore`
- Set up folder structure: `src/`, `tests/`, `data/`, `docs/`
- Write initial README with project vision

✓ **Deliverable:** Dev environment ready, 30 priority formulas identified, Git repo initialized

Week 1 Summary

Total Hours: 9 hours

Key Outcomes: Market validation initiated, technical foundation established, ready for development

Next Week Preview: Build core discovery engine and integrate with LLM

Week 2: MVP Development - Discovery Engine

Dates: February 3-7

Working Days: Monday, Tuesday, Thursday, Friday, Saturday

Total Hours: 15 hours

Objective

Build hybrid discovery system (symbolic regression + LLM interpretation).

Day 4 - Monday, February 3 (3 hours)

Focus: Symbolic Regression Engine

Hour 1 (9:00-10:00): PySR wrapper class

- Create src/symbolic_engine.py
- Implement SymbolicDiscovery class with configurable parameters
- Add physics-informed constraints (positive values, bounded ranges)

Hour 2 (10:00-11:00): Test on simple cases

- Test Case 1: Linear relationship ($y = ax + b$)
- Test Case 2: Impermanent Loss formula (known formula)
- Verify PySR discovers correct expressions

Hour 3 (11:00-12:00): DeFi-specific constraints

- Add reserve constraints ($x > 0, y > 0$)
- Add fee bounds ($\text{fee} \in [0, 1]$)
- Add price impact constraints ($\text{output} < \text{input}$)

 **Deliverable:** Working symbolic regression engine with DeFi constraints

Day 5 - Tuesday, February 4 (3 hours)

Focus: LLM Integration

Hour 1 (9:00-10:00): Claude API wrapper

- Create src/llm_interpreter.py
- Implement LLMInterpreter class with error handling
- Design domain-specific prompts for DeFi and Risk formulas

Hour 2 (10:00-11:00): Prompt engineering

- Create prompt template: formula — interpretation + use cases
- Test with 5 known formulas (IL, VaR, Sharpe ratio, etc.)
- Refine prompts based on quality of interpretations

Hour 3 (11:00-12:00): Response parsing

- Implement JSON parsing for structured LLM responses
- Add validation for required fields (interpretation, params, assumptions)
- Error handling for malformed API responses

✓ **Deliverable:** LLM interpreter working, producing quality interpretations

Day 6 - Thursday, February 6 (3 hours)

Focus: Validation Framework - Layer 1 & 2

Hour 1 (9:00-10:00): Symbolic validator (Layer 1)

- Create src/validators/symbolic_validator.py
- Use SymPy to parse discovered formulas
- Check for: undefined variables, invalid operations, simplification

Hour 2 (10:00-11:00): Dimensional validator (Layer 2)

- Create src/validators/dimensional_validator.py
- Use Pint for unit checking
- Verify: price * quantity = value, rate * time = amount

Hour 3 (11:00-12:00): Test validation layers

- Create test suite with 10 formulas (5 valid, 5 invalid)
- Verify symbolic validator catches syntax errors
- Verify dimensional validator catches unit mismatches

✓ **Deliverable:** 2 validation layers working, catching 80%+ of errors

Day 7 - Friday, February 7 (3 hours)

Focus: Domain Validators & Ensemble Scoring

Hour 1 (9:00-10:00): Domain validator (Layer 3)

- Create src/validators/domain_validator.py
- DeFi rules: reserves > 0, fees $\in [0,1]$, reasonable slippage
- Risk rules: probabilities $\in [0,1]$, VaR < 0, realistic confidence intervals

Hour 2 (10:00-11:00): Ensemble validator (Layer 4)

- Create src/validators/ensemble_validator.py
- Weighted scoring: symbolic 25%, dimensional 25%, domain 30%, numerical 20%
- Overall pass threshold: 75/100 score

Hour 3 (11:00-12:00): Integration testing

- Test end-to-end: discover $\rightarrow$ validate (all 4 layers) $\rightarrow$ score
- Verify ensemble catches edge cases missed by individual layers
- Create validation report template

✓ **Deliverable:** Complete 4-layer validation framework, ensemble scoring working

Day 8 - Saturday, February 8 (3 hours)

Focus: Hybrid System Integration

Hour 1 (9:00-10:00): Create hybrid system

- Create src/hybrid_system.py
- Integrate: symbolic_engine + validators + llm_interpreter
- Implement full pipeline: data → discovery → validation → interpretation

Hour 2 (10:00-11:00): End-to-end testing

- Test with Impermanent Loss formula (known ground truth)
- Test with VaR calculation (risk domain)
- Verify all components work together

Hour 3 (11:00-12:00): Performance optimization

- Profile bottlenecks (likely PySR discovery time)
- Add caching for repeated validations
- Optimize LLM prompts for token efficiency

✓ **Deliverable:** Fully integrated hybrid system, ready for production formulas

Week 2 Summary

Total Hours: 15 hours

Key Outcomes: Core discovery engine built, LLM integrated, 2 validation layers operational

Next Week Preview: Complete validation framework, generate first 30 formulas

Week 3: MVP Development - Validation & Integration

Dates: February 10-14

Working Days: Monday, Tuesday, Thursday, Friday, Saturday

Total Hours: 15 hours

Objective

Generate and validate 30+ DeFi/Risk formulas, create demo interface.

Day 9 - Monday, February 10 (3 hours)

Focus: DeFi Formula Generation (Batch 1)

Hour 1 (9:00-10:00): Core DeFi formulas

- Generate: Impermanent Loss (IL), Constant Product AMM, Price Impact
- Run through validation pipeline
- Target: 5 formulas validated

Hour 2 (10:00-11:00): Liquidity & fees

- Generate: LP position sizing, Fee APR, Utilization rate
- Validate and interpret
- Target: 5 more formulas (total: 10)

Hour 3 (11:00-12:00): Analysis & refinement

- Review validation scores, identify patterns in failures
- Refine domain constraints based on results
- Document learnings for next batch

 **Deliverable:** 10 validated DeFi formulas with interpretations

Day 10 - Tuesday, February 11 (3 hours)

Focus: Risk Formula Generation (Batch 1)

Hour 1: Core risk metrics

- VaR 95%, Expected Shortfall (CVaR), Sharpe Ratio

Hour 2: Portfolio risk

- Portfolio variance, correlation matrix, beta calculation

Hour 3: Validate batch

- Run all through validation, aim for 85%+ pass rate
- Target: 10 risk formulas (total: 20)

Day 11 - Thursday, February 13 (3 hours)

Focus: Advanced Formulas (Batch 2)

Hour 1: DeFi advanced

- Multi-hop swaps, concentrated liquidity, time-weighted IL

Hour 2: Risk advanced

- Multi-asset VaR, skewness-adjusted CVaR, tail risk

Hour 3: Validation round

- Validate new batch, refine failing formulas
- Target: 30 total formulas

Day 12 - Friday, February 14 (3 hours)

Focus: Demo Dashboard Creation

Hour 1: Streamlit app setup

- Create demo_app.py, basic layout with sidebar

Hour 2: Interactive features

- Formula discovery interface, validation visualization

Hour 3: Pre-load examples

- Add 10 best examples, polish UI/UX

Day 13 - Saturday, February 15 (3 hours)

Focus: Testing & Documentation

Hour 1: Integration testing

- Test full workflow with 5 new formulas

Hour 2: Create documentation

- API docs, usage examples, README improvements

Hour 3: Performance metrics

- Calculate: avg discovery time, validation rate, cost per formula

Week 3 Summary

Total Hours: 15 hours

Key Outcomes: 30+ formulas validated, demo dashboard created, ready for customer presentation

Next Week Preview: Polish demo, conduct pilot testing, prepare launch materials

Week 4: Demo, Testing & Launch Preparation

Dates: February 17-21

Working Days: Monday, Tuesday, Thursday, Friday, Saturday

Total Hours: 15 hours

Objective

Polish demo, conduct pilot customer testing, prepare launch materials.

Day 14 - Monday, February 17 (3 hours)

Focus: Demo Polish & Video

Hour 1: UI improvements

- Enhance dashboard visuals, add charts for validation scores

Hour 2: Record demo video

- 5-minute walkthrough: problem → solution → results

Hour 3: Create pitch deck

- 10 slides: problem, solution, demo, traction, ask

Day 15 - Tuesday, February 18 (3 hours)

Focus: Customer Discovery Calls

Hour 1: Prepare for calls

- Review protocols researched in Week 1, prepare demo flow

Hour 2: Conduct 2 discovery calls

- Present demo, gather feedback, identify pain points

Hour 3: Document insights

- Synthesize feedback, update product roadmap

Day 16 - Thursday, February 20 (3 hours)

Focus: Pilot Testing

Hour 1: Set up pilot

- Work with 1-2 protocols to test system on their data

Hour 2: Monitor usage

- Track: formulas generated, validation rates, errors

Hour 3: Gather feedback

- Interview pilot users, identify improvements

Day 17 - Friday, February 21 (3 hours)

Focus: Paper Draft & Results

Hour 1: Start paper outline

- Abstract, Introduction, Methodology sections

Hour 2: Results analysis

- Calculate statistics: success rate, time savings, accuracy

Hour 3: Create figures

- 3 key charts: validation scores, comparison vs baselines, cost analysis

Day 18 - Saturday, February 22 (3 hours)

Focus: Launch Preparation

Hour 1: GitHub repo polish

- README, examples, contribution guide

Hour 2: Marketing materials

- Twitter thread, LinkedIn post, blog post draft

Hour 3: Pricing strategy

- Define tiers: \$500-2k/month for protocols, \$10k+ enterprise

Week 4 Summary

Total Hours: 15 hours

Key Outcomes: Demo ready, 2+ pilot customers, launch materials complete

Next Week Preview: Final polish based on feedback, prepare for wider launch

Week 5: Polish & Customer Outreach

Dates: February 24-26

Working Days: Monday, Tuesday

Total Hours: 6 hours

Objective

Incorporate pilot feedback, finalize MVP, launch to broader audience.

Day 19 - Monday, February 24 (3 hours)

Focus: Feedback Integration

Hour 1: Review all feedback

- Prioritize improvements by impact and effort

Hour 2: Critical fixes

- Address 2-3 high-impact issues from pilots

Hour 3: Final testing

- Run through all 30+ formulas, ensure 85%+ validation

Day 20 - Tuesday, February 25 (3 hours)

Focus: Launch Day

Hour 1: Final polish

- Update README, ensure all docs current, test demo one last time

Hour 2: Go public

- Post to Twitter, LinkedIn, relevant communities
- Email 20 protocols with demo link

Hour 3: Monitor & respond

- Track signups, respond to questions, collect feedback

Week 5 Summary

Total Hours: 6 hours

Key Outcomes: MVP refined, launched publicly, initial customer traction

MVP COMPLETE - Decision Point

After Week 5, evaluate:

- ✅ If 3+ protocols show strong interest → **Full commitment to DeFi/Risk product**
- ⚠️ If lukewarm response → **Pivot to ANREPORT, use DeFi as one vertical**

Either way, you'll have: MVP ✓ | Paper draft ✓ | Market validation data ✓

Appendix

Success Metrics

Metric	Minimum Viable	Target
Validated Formulas	30	50
Validation Success Rate	80%	85%
Pilot Customer Interest	2	3+

Risk Mitigation

- **Technical Risk: Integration complexity:** Start with simpler validations, iterate to complexity
- **Market Risk: Lukewarm customer response:** Have ANREPORT pivot ready; core tech works for both
- **Time Risk: Underestimating task duration:** Built in 20% buffer; minimum viable metrics defined
- **Resource Risk: API costs:** Budget \$200 for Claude API; optimize prompts early

Next Steps After Week 4

1. Week 5-8: Incorporate feedback, refine product-market fit
2. Week 9-12: First paying customers, revenue validation
3. Week 13-16: Scale to 5-10 customers, prove business model
4. Week 17+: Expand to ANREPORT vision with proven foundation